

NAME:

NO:

CLASS:

RIVERSIDE SECONDARY SCHOOL



PRELIMINARY EXAMINATION 2026

SUBJECT : ADDITIONAL MATHEMATICS

SUBJECT CODE /PAPER : 4049/01

LEVEL/STREAM : Sec 4 Express

DURATION : 2 hours 15 minutes

Candidates answer on the Question Paper.

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures or in 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 90.

Mathematical Formulae

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Binomial expansion

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{r}a^{n-r}b^r + \dots + b^n,$$

where n is positive integer and $\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n(n-1)\dots(n-r+1)}{r!}$

2. TRIGONOMETRY

Identities

$$\sin^2 A + \cos^2 A = 1$$

$$\sec^2 A = 1 + \tan^2 A$$

$$\operatorname{cosec}^2 A = 1 + \cot^2 A$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$

$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$$

$$\sin 2A = 2 \sin A \cos A$$

$$\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A$$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\Delta = \frac{1}{2} bc \sin A$$

Answer all questions.

- 1 Given that $(2 + \sqrt{5})^2 - \frac{22}{4 - \sqrt{5}} = p + q\sqrt{5}$, where p and q are integers, find, without using a calculator, the value of p and of q . [4]

$$\begin{aligned}
 & (2 + \sqrt{5})^2 - \frac{22}{4 - \sqrt{5}} \\
 &= 4 + 4\sqrt{5} + 5 - \frac{22}{4 - \sqrt{5}} \cdot \frac{4 + \sqrt{5}}{4 + \sqrt{5}} \\
 &= 9 + 4\sqrt{5} - \frac{88 + 22\sqrt{5}}{16 - 5} \\
 &= \frac{99 + 44\sqrt{5} - 88 - 22\sqrt{5}}{11} \\
 &= \frac{11 + 22\sqrt{5}}{11}
 \end{aligned}$$

$$p = 1, q = 2$$

- 2 Given that $y = \ln\left(\frac{3x+1}{x-2}\right)$, $x > 2$, prove that y decreases for all real values of x . [4]

$$y = \ln(3x+1) - \ln(x-2)$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{3}{3x+1} - \frac{1}{x-2} \\ &= \frac{3(x-2) - (3x+1)}{(3x+1)(x-2)} \\ &= \frac{3x-6-3x-1}{(3x+1)(x-2)} \\ &= \frac{-7}{(3x+1)(x-2)}\end{aligned}$$

For all real values of x , $x > 2$

$$x-2 > 0,$$

$$3x+1 > 0,$$

$$-7 < 0,$$

$$\frac{-7}{(3x+1)(x-2)} < 0$$

$$\frac{dy}{dx} < 0. \text{ Hence, } y \text{ decreases.}$$

- 3 Find the equation of the normal to the curve $y = 2\sin 3x + \cos^4(2x)$ at the point Q when $x = 0$. [6]

$$y = 2\sin 3x + \cos^4(2x)$$

$$\frac{dy}{dx} = 2\cos 3x \bullet 3 + 4\cos^3(2x) \bullet -\sin(2x) \bullet 2$$

$$= 6\cos 3x - 8\cos^3(2x)\sin(2x)$$

$$x = 0, y = 1$$

$$\frac{dy}{dx} = 6$$

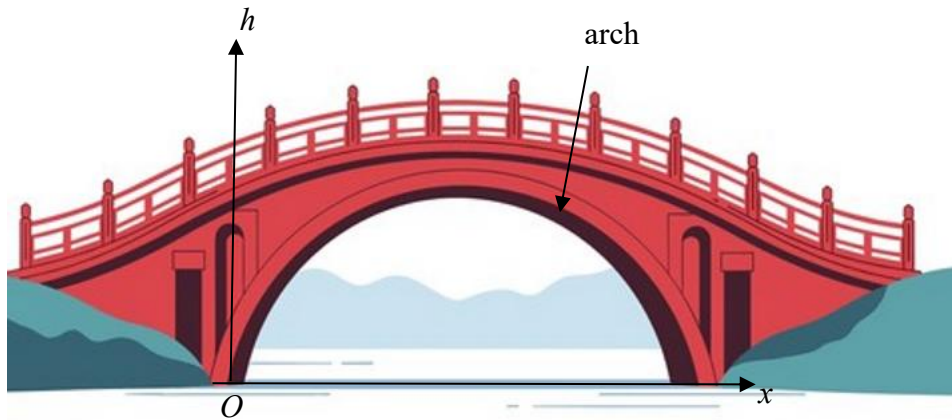
$$m_{normal} = -\frac{1}{6}$$

$$y - 1 = -\frac{1}{6}(x - 0)$$

$$y = -\frac{1}{6}x + 1$$

- 4 China has many arch bridges which are mainly used to bridge valleys, gorges and rivers. The arch bridge also allows boat to pass through and is a form of transportation.

The arch at the bottom of the bridge below is modelled by a function $h = -0.1x^2 + 4x$, where h is the vertical distance, in metres, from the water surface and x is the horizontal distance, in metres, from start of the arch point O .



- (a) Write down the maximum height of the arch. [2]

$$\begin{aligned}
 h &= -0.1x^2 + 4x \\
 &= -0.1[(x - 20)^2 - 400] \\
 &= -0.1(x - 20)^2 + 40 \\
 \text{max height} &= 40\text{m}
 \end{aligned}$$

- (b) A large boat requires a vertical clearance of at least 30 metres to safely cruise under the bridge. Solve $-0.1x^2 + 4x \geq 30$ to find the range of values of x in which the boat can safely pass through. [2]

$$\begin{aligned}
 -0.1x^2 + 4x &\geq 30 \\
 -x^2 + 40x - 300 &\geq 0 \\
 x^2 - 40x + 300 &\leq 0 \\
 (x - 30)(x - 10) &\leq 0 \\
 10 &\leq x \leq 30
 \end{aligned}$$

- 5** Water is poured into a hemispherical bowl. The height of the water level is h cm. The volume of water in the bowl, $V \text{ cm}^3$, varies with h and is given by

$$V = \pi \int_0^h 400 - (x - 20)^2 dx.$$

- (a)** Show that $V = \frac{\pi}{3}(60h^2 - h^3)$. [3]

$$\begin{aligned} V &= \pi \int_0^h 400 - (x - 20)^2 dx \\ &= \pi \int_0^h 400 - (x^2 - 40x + 400) dx \\ &= \pi \int_0^h (-x^2 + 40x) dx \\ &= \pi \left[-\frac{x^3}{3} + 20x^2 \right]_0^h \\ &= \pi \left[-\frac{h^3}{3} + 20h^2 \right] \\ &= \frac{\pi}{3}(60h^2 - h^3) \end{aligned}$$

- (b)** Water is poured into the bowl at a constant rate of $30 \text{ cm}^3/\text{s}$.
Find the rate of increase of the height of the water level when $h = 8$, expressing your answer in terms of π . [3]

$$\begin{aligned} \frac{dV}{dt} &= 30 \\ \frac{dV}{dh} &= \frac{\pi}{3}(120h - 3h^2) = \pi(40h - h^2) \\ \text{when } h=8, \frac{dV}{dh} &= 256\pi \\ \frac{dV}{dt} &= \frac{dV}{dh} \times \frac{dh}{dt} \\ 30 &= 256\pi \times \frac{dh}{dt} \\ \frac{dh}{dt} &= \frac{30}{256\pi} = \frac{15}{128\pi} \end{aligned}$$

- 6 (a) Explain why $x^2 + ax + a^2 + 2$ is always positive for all real values of x . [3]

$$\begin{aligned}
 & x^2 + ax + a^2 + 2 \\
 & b^2 - 4ac \\
 & = (a)^2 - 4(1)(a^2 + 2) \\
 & = a^2 - 4a^2 - 8 \\
 & = -3a^2 - 8
 \end{aligned}$$

For all real values of a , $-3a^2 \leq 0$, $-3a^2 - 8 \leq -8 < 0$

Since coefficient of $x^2 > 0$, it will be an opening upwards curve. $x^2 + ax + a^2 + 2$ is always positive for all real values of a .

- (b) Find the set of values of m for which the line $y = mx - 4$ intersects the curve $y = -3x^2 + 2x - 7$ at two distinct points. [4]

$$\begin{aligned}
 mx - 4 &= -3x^2 + 2x - 7 \\
 -3x^2 + x(2 - m) - 3 &= 0 \\
 b^2 - 4ac &> 0 \\
 (2 - m)^2 - 4(-3)(-3) &> 0 \\
 4 - 4m + m^2 - 36 &> 0 \\
 m^2 - 4m - 32 &> 0 \\
 (m - 8)(m + 4) &> 0 \\
 m < -4 \quad \text{or} \quad m > 8
 \end{aligned}$$

7 (a) Prove that $\frac{\cot \theta - \tan \theta}{\cot \theta + \tan \theta} = \cos 2\theta$ [3]

$$\begin{aligned}
 & \frac{\cot \theta - \tan \theta}{\cot \theta + \tan \theta} \\
 &= \left(\frac{\cos \theta}{\sin \theta} - \frac{\sin \theta}{\cos \theta} \right) \div \left(\frac{\cos \theta}{\sin \theta} + \frac{\sin \theta}{\cos \theta} \right) \\
 &= \left(\frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta} \right) \div \left(\frac{\cos^2 \theta + \sin^2 \theta}{\sin \theta \cos \theta} \right) \\
 &= \left(\frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta \cos \theta} \right) \times \left(\frac{\sin \theta \cos \theta}{1} \right) \\
 &= \cos^2 \theta - \sin^2 \theta \\
 &= \cos 2\theta
 \end{aligned}$$

(b) Hence solve $1 + \frac{2 \cot \theta - 2 \tan \theta}{\cot \theta + \tan \theta} = 0$ for $-\pi < \theta < \pi$. [4]

$$\begin{aligned}
 1 + \frac{2 \cot \theta - 2 \tan \theta}{\cot \theta + \tan \theta} &= 0 \\
 1 + 2 \left(\frac{\cot \theta - \tan \theta}{\cot \theta + \tan \theta} \right) &= 0 \\
 1 + 2 \cos 2\theta &= 0 \\
 \cos 2\theta &= -\frac{1}{2} \\
 \alpha &= 1.0472 \\
 2\theta &= -(\pi - \alpha), -(\pi + \alpha), \pi - \alpha, \pi + \alpha \\
 \theta &= -1.05, -2.09, 1.05, 2.09
 \end{aligned}$$

- 8 A marine biologist is studying how the intensity of light, I lux, decreases at a depth of d metres below the surface of the sea. It is believed that I and d can be modelled by the formula $I = Ae^{-kd}$, where A and k are constants. The table below shows corresponding values of I and d .

d	2	4	6	8	10
I	445	198	88	40	18

- (a) Explain how a straight line graph can be drawn to represent the formula and how the value of k can be obtained from the line. [2]

$$I = Ae^{-kd}$$

$$\ln I = \ln A + (-kd) \ln e$$

$$Y = \ln I$$

$$X = d$$

$$c = \ln A$$

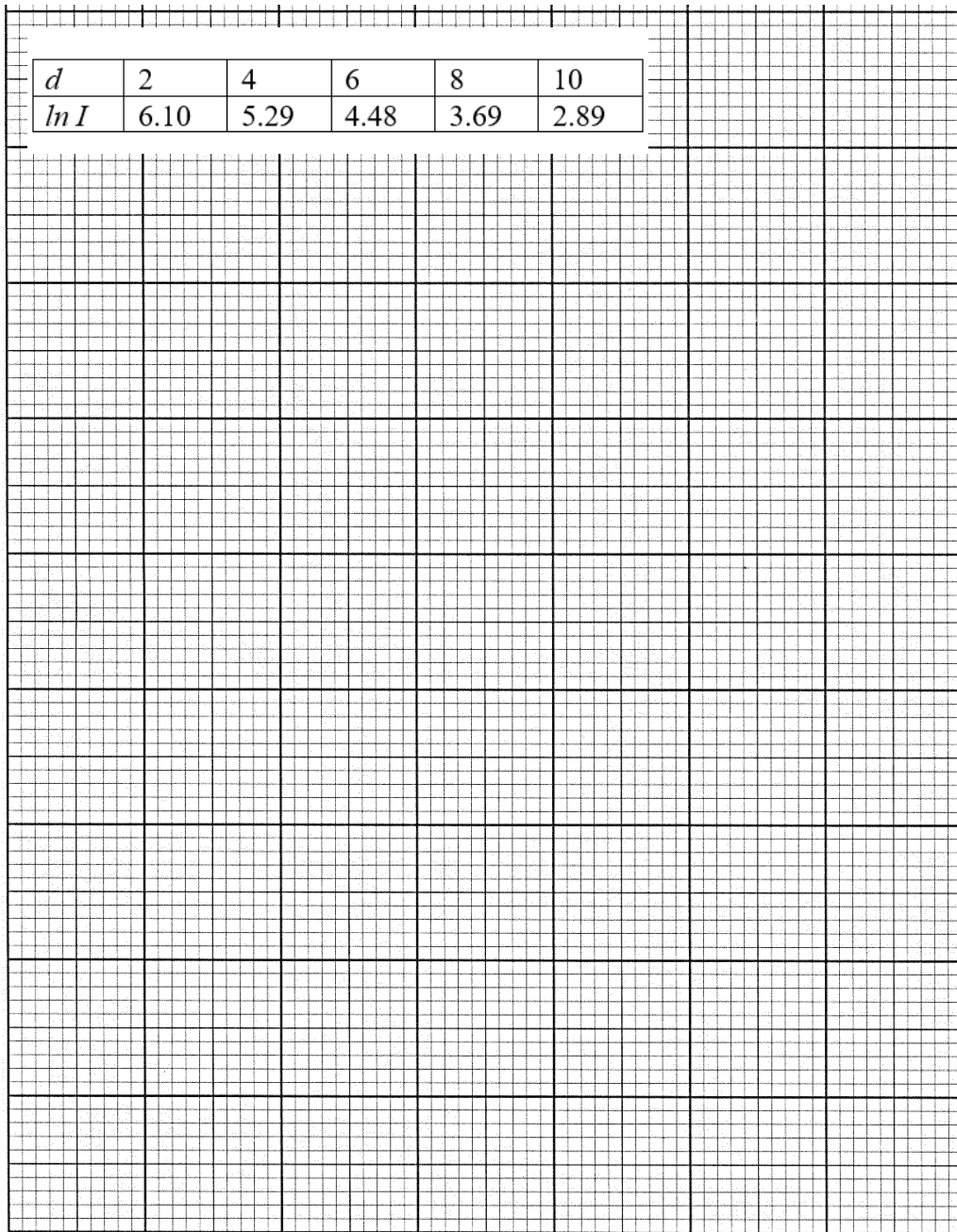
$$m = -k$$

Plot $\ln I$ as the vertical axis, d as the horizontal axis.

Calculate the gradient. $k = -m$

(b) Plot $\ln I$ against d to illustrate the information.

[2]



(c) Use your graph to show that the estimated surface intensity when $d = 0$ is

1000 lux.

[2]

$$\ln A = 6.9$$

$$A = e^{6.9} = 992 = 1000$$

- (d) The photic zone is defined as the depth where light intensity is at least 1% of the surface intensity. Use your graph to estimate the maximum depth of the photic zone. [2]

$$1000 \times 1\% = 10$$

$$\ln 10 = 2.3$$

$$11.5m$$

- 9 (a) Express $\frac{x^2 + x + 6}{(x^2 + 3)(1 - x)}$ as a sum of $\frac{A}{(1 - x)}$ and $\frac{Bx + C}{x^2 + 3}$. [5]

$$\frac{x^2 + x + 6}{(x^2 + 3)(1 - x)} = \frac{A}{(1 - x)} + \frac{Bx + C}{x^2 + 3}$$

$$x^2 + x + 6 = A(x^2 + 3) + (Bx + C)(1 - x)$$

$$\text{Let } x = 1$$

$$8 = 4A$$

$$A = 2$$

$$\text{Let } x = 0$$

$$6 = 3A - C$$

$$C = 0$$

$$\text{Let } x = 2$$

$$12 = 14 - 2B$$

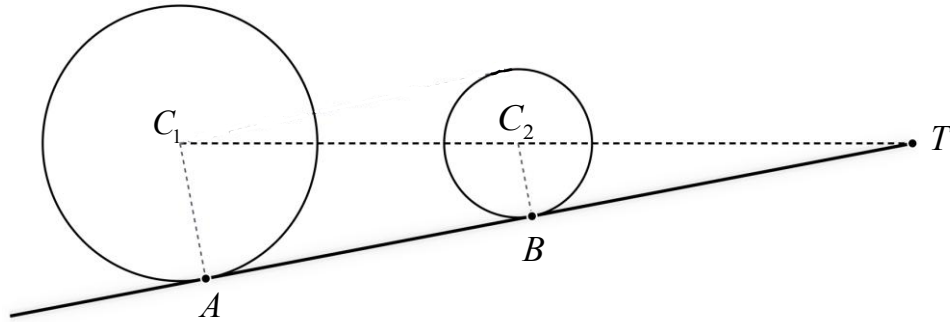
$$B = 1$$

$$\frac{x^2 + x + 6}{(x^2 + 3)(1 - x)} = \frac{2}{(1 - x)} + \frac{x}{x^2 + 3}$$

- (b) Hence evaluate $\int_{-2}^{-1} \frac{x^2 + x + 6}{(x^2 + 3)(1 - x)} dx$, expressing your answer in the form of $\ln\left(\frac{a}{b\sqrt{c}}\right)$.
[4]

$$\begin{aligned}
 & \int_{-2}^{-1} \frac{x^2 + x + 6}{(x^2 + 3)(1 - x)} dx \\
 &= \int_{-2}^{-1} \frac{2}{(1 - x)} + \frac{x}{x^2 + 3} dx \\
 & \left[-2 \ln(1 - x) + \frac{1}{2} \ln(x^2 + 3) \right]_{-2}^{-1} \\
 &= \left[-2 \ln(2) + \frac{1}{2} \ln(4) \right] - \left[-2 \ln(3) + \frac{1}{2} \ln(7) \right] \\
 &= \ln\left(\frac{1}{4}\right) + \ln 2 + \ln 9 - \ln \sqrt{7} \\
 &= \ln\left(\frac{9}{2\sqrt{7}}\right)
 \end{aligned}$$

- 10** ABT is a common tangent to the big circle A and small circle B whose centres are C_1 and C_2 respectively. Points A and B are on the circumference of the big and small circle respectively. The line joining C_1 and C_2 meets tangent ABT at point T .



- (a)** The equation of the small circle B is $x^2 + y^2 - 14x + 45 = 0$.

Find the radius and coordinates of the centre of the circle.

[3]

$$x^2 + y^2 - 14x + 45 = 0$$

$$(x-7)^2 - 49 + y^2 = -45$$

$$(x-7)^2 + y^2 = 2^2$$

$$(7, 0), \text{ radius} = 2$$

- (b)** Circle A has a radius of 4 and C_1 lies on the origin.

Write down the equation of circle A .

[1]

$$x^2 + y^2 = 16$$

- (c) Write down the coordinates of point T.

[2]

(14, 0)

- (d) Determine if $X(2, 3)$ lies inside, outside or on circle A .

[2]

$$C_1X = \sqrt{2^2 + 3^2}$$

$$= \sqrt{13}$$

Since $OX = \sqrt{13} < 4$, X lies inside the circle.

11 The equation of a curve is $y = x^2(e^{-x})$.

- (a) Show that $\frac{dy}{dx} = \frac{x(2-x)}{e^x}$ [2]

$$\begin{aligned} y &= \frac{x^2}{e^x} \\ \frac{dy}{dx} &= \frac{e^x(2x) - x^2(e^x)}{e^{2x}} \\ &= \frac{e^x(2x - x^2)}{e^{2x}} \\ &= \frac{x(2-x)}{e^x} \end{aligned}$$

- (b) Find the range of values of x such that $\frac{dy}{dx} > 0$.

Describe the nature of this function within this range. [2]

$$x(2-x) > 0$$

$$0 < x < 2$$

Increasing function

(c) Show that $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = \frac{2}{e^x}$. [4]

$$\frac{dy}{dx} = \frac{x(2-x)}{e^x}$$

$$\frac{d^2y}{dx^2} = \frac{e^x(2-2x) - (2x-x^2)e^x}{e^{2x}}$$

$$= \frac{e^x(2-2x-2x+x^2)}{e^{2x}}$$

$$= \frac{x^2 - 4x + 2}{e^x}$$

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = \frac{2}{e^x}$$

LHS

$$= \frac{x^2 - 4x + 2}{e^x} + 2\left(\frac{x(2-x)}{e^x}\right) + \frac{x^2}{e^x}$$

$$= \frac{x^2 - 4x + 2 + 4x - 2x^2 + x^2}{e^x}$$

$$= \frac{2}{e^x}$$

= RHS

- 12 A teenager is testing a toy car on a long driveway. The car passes through a fixed chalk line, O . Its velocity, v m/s, at time t seconds after leaving O , is given by

$$v = 4t^2 - 24t + 32.$$

- (a) Show that the car is instantaneously at rest when $t = 2$ and $t = 4$. [2]

$$0 = 4t^2 - 24t + 32$$

$$t^2 - 6t + 8 = 0$$

$$(t - 4)(t - 2) = 0$$

$$t = 2, \quad t = 4$$

- (b) Find the minimum velocity of the car. $\frac{dv}{dt} = 8t - 24$ [3]

$$\frac{d^2v}{dt^2} = 8 > 0$$

$$t = 3$$

Hence, min value

$$v = 4(3)^2 - 24(3) + 32$$

$$= -4 \text{ m/s}$$

- (c) Find an expression, in terms of t , for the distance of the toy from O at time t . [2]

$$s = \int 4t^2 - 24t + 32 \, dt$$

$$= \frac{4}{3}t^3 - 12t^2 + 32t + c$$

$$t = 0, s = 0, c = 0$$

$$s = \frac{4}{3}t^3 - 12t^2 + 32t$$

- (d) Find the distance travelled at the third second. [2]

$$t = 2, s = \frac{80}{3}$$

$$t = 3, s = 24$$

$$\text{distance} = \frac{80}{3} - 24 = 2.67m$$

- (e) Find the total distance travelled by the car in the first 5 seconds. [2]

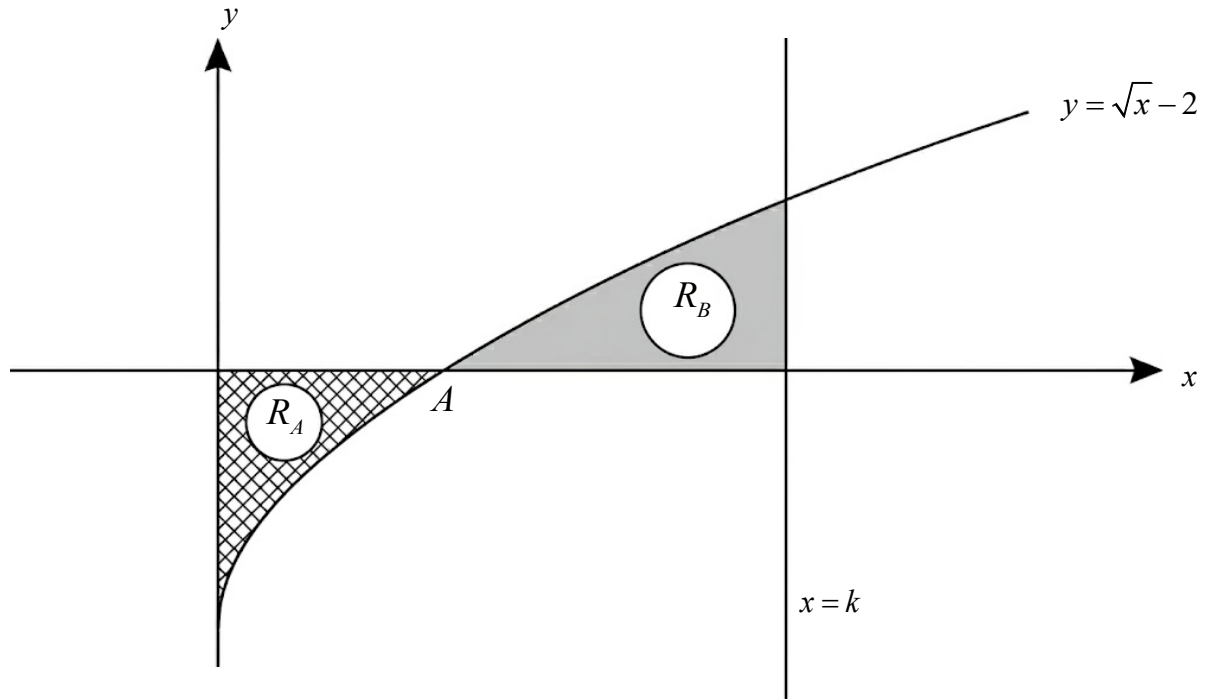
$$t = 4, s = \frac{64}{3}$$

$$t = 5, s = \frac{80}{3}$$

$$\begin{aligned} \text{distance} &= \frac{80}{3} + \left(\frac{80}{3} - \frac{64}{3} \right) + \left(\frac{80}{3} - \frac{64}{3} \right) \\ &= 37.3m \end{aligned}$$

- 13 The diagram shows the curve $y = \sqrt{x} - 2$ and the line $x = k$, where k is a constant, $k > 4$. The curve intersects the x -axis at point A .

R_A is area of region bounded by the curve, y -axis and x -axis. R_B is the area of region bounded by the curve, x -axis and vertical line $x = k$.



- (a) Show that the x -coordinate of A is 4.

[2]

$$0 = \sqrt{x} - 2$$

$$\sqrt{x} = 2$$

$$x = 2^2$$

$$x = 4$$

Area of region A , R_A , and region B , R_B , are equal.

(b) Find the value of k .

[6]

$$\begin{aligned}
 & \int_0^4 \sqrt{x} - 2 \, dx \\
 &= \int_0^4 x^{\frac{1}{2}} - 2 \, dx \\
 &= \left[\frac{x^{1.5}}{1.5} - 2x \right]_0^4 \\
 &= \left[\frac{4^{1.5}}{1.5} - 2(4) \right] \\
 &= -\frac{8}{3} \\
 R_A &= \frac{8}{3} \\
 & \int_4^k x^{\frac{1}{2}} - 2 \, dx \\
 &= \left[\frac{x^{1.5}}{1.5} - 2x \right]_4^k \\
 &= \left[\frac{k^{1.5}}{1.5} - 2k \right] - \left[\frac{4^{1.5}}{1.5} - 2(4) \right]
 \end{aligned}$$

$$\frac{8}{3} = \frac{2}{3}k^{1.5} - 2k + \frac{8}{3}$$

$$\frac{2}{3}k^{1.5} - 2k = 0$$

$$2k\left(\frac{1}{3}k^{0.5} - 1\right) = 0$$

$$\frac{1}{3}k^{0.5} - 1 = 0$$

$$\frac{1}{k^{\frac{1}{2}}} = 3$$

$$k = 9$$